SE-JEPA: Self-Evolving Latent World Models

Anonymous CVPR submission

Abstract

Latent world models let agents plan by imagining the outcomes of their actions, but they stop learning once deployed: when objects or appearances shift, prediction errors become planning failures. Test-time adaptation such as AdaJEPA repairs these errors within an episode but then resets, meeting every recurring shift as if for the first time. We therefore seek a world model that remembers what it learns. However, we find that simply keeping the adapted weights fails: updates across episodes are nearly orthogonal, and accumulating them degrades planning. To overcome this, we present SE-JEPA, a self-evolving latent world model. Inspired by complementary learning systems, it pairs fast within-episode adaptation with slow cross-episode consolidation, replaying experience into slow learners, one per self-discovered domain. Supervised only by its own prediction errors, SE-JEPA decides what to consolidate, where to store it, and when to trust it for planning. Because its pretrained model is never overwritten, nothing it already knows can be forgotten. It requires no shift labels or task boundaries and benefits both gradient-based and sampling-based planners. Across unseen object shapes, visual shifts, and deformable rope and granular manipulation, SE-JEPA grows stronger the longer it is deployed, setting a new state of the art by a wide margin.

1. Introduction

World models let an agent try actions in imagination before committing to them. Latent world models make this practical for visual control: they predict how an encoded scene evolves under candidate actions, and a model-predictive controller (MPC) searches for the action sequence whose predicted outcome matches an image goal [21, 35, 40]. Planning is only as good as these predictions. When a deployed agent meets an object shape or a visual appearance it was not trained on, prediction errors turn directly into planning failures.

Test-time adaptation offers a remedy. AdaJEPA [34] treats every executed action as free supervision: the observed transition corrects the predictor before the next replan. This closes a second feedback loop around MPC and recovers much of the lost performance within an episode. The correction, however, is discarded when the episode ends. An agent that pushes the same unfamiliar object a hundred times relearns its dynamics a hundred times.

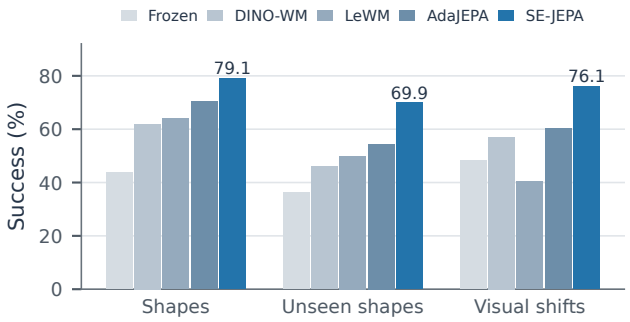


Figure 1. **Deployment experience improves held-out planning.** Held-out success with CEM planning on PushObj shapes, the three shapes unseen during pretraining, and six PushT visual shifts.

We ask whether a world model can instead *remember* what it learns during deployment. The key is to turn local corrections into reusable predictions. A fast update fits a short trajectory around one start and goal; across episodes, these updates point in nearly orthogonal directions. Directly retaining them does not reliably produce a predictor that serves the full stream. On a matched stream, full accumulation lowers success from 62% to 48% (Sec. 3.1). What should persist is the experience behind an update, not the update itself.

We introduce SE-JEPA, a self-evolving latent world model built on complementary learning systems [22]. A fast system adapts within the episode exactly as AdaJEPA does. A slow system stores the episode's transitions in the fixed latent space of the pretrained encoder and consolidates them into slow predictors by replay, one per domain that the model discovers on its own. Prediction errors and fast-update fingerprints organize this memory without domain supervision. The error of the pretrained predictor tells SE-JEPA whether an episode contains anything new. The direction of the fast update tells it which domain the experience belongs to. The error of a slow predictor on episodes it has not yet trained on tells it when that predictor can be trusted for planning. The pretrained model is never modi-

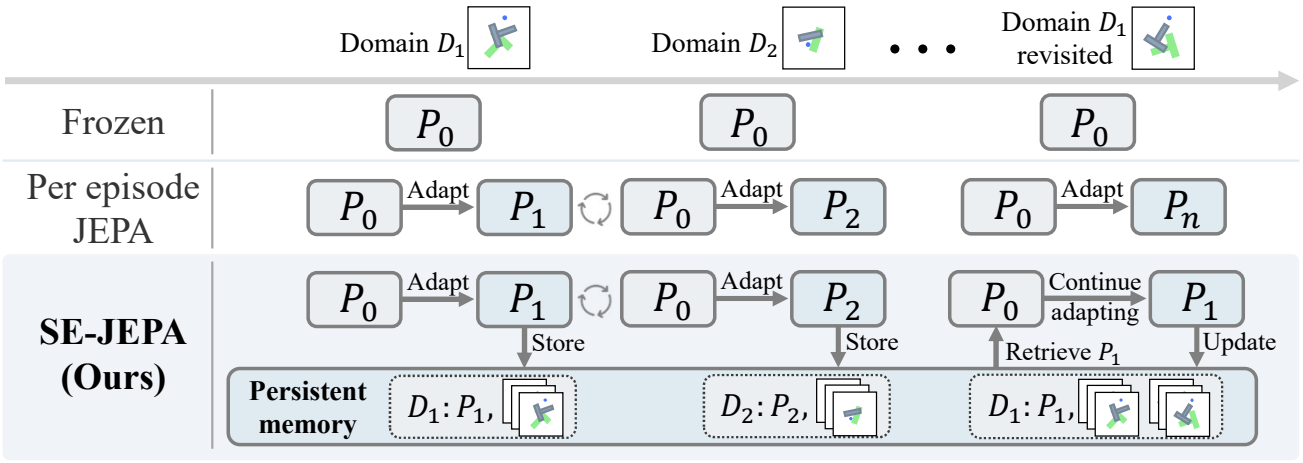


Figure 2. **From episodic adaptation to self-evolution.** A frozen world model plans every episode with the same predictor. Single-episode adaptation repairs the predictor during an episode and resets it afterwards. SE-JEPA consolidates experience into slow learners by replay and retrieves them when a shift recurs, so experience from earlier episodes can improve later planning. Domain names are for illustration; the method receives no labels.

fied, so the agent can always fall back to it.

This design connects continual learning directly to visual control. With both gradient descent (GD) and the cross-entropy method (CEM), SE-JEPA improves held-out planning on PushObj shapes and PushT visual conditions (Fig. 1). Slow consolidation also improves rope and granular manipulation, extending the benefit to deformable dynamics. With CEM it reaches 79.1% success on held-out shapes and 69.9% on unseen shapes, against 70.4% and 54.4% for AdaJEPA; under recoloring of the object or goal its gains exceed 27 points. Its online success keeps rising across deployment rounds, and on deformable objects it reduces Chamfer distance by 49% (rope) and 27% (granular) relative to DINO-WM.

Our contributions are:

- We identify experience replay as the bridge from episodic adaptation to persistent latent planning, motivated by a matched-stream diagnosis of weight accumulation.
- We propose SE-JEPA, which turns transient adaptation into lasting predictive knowledge through replay consolidation, self-discovered domains, and validation before retrieval, using prediction errors and fast-update fingerprints.
- We demonstrate higher held-out success and fewer interactions on shape and visual shifts with two planners, together with improved deformable manipulation through slow consolidation.

2. Related Work

World models for control. Learned dynamics support control by predicting the consequences of candidate actions. Embed to Control [36] learns locally linear latent dynamics, World Models [8] learns recurrent simulators, and PETS [6] combines probabilistic ensembles with planning. PlaNet [9] plans in latent space; Dreamer and its successors learn behavior through latent imagination [10–12]. SimPLe [15] and IRIS [23] learn visual simulators for data-efficient policy learning. MuZero [28] and EfficientZero [37] learn models tailored to search, while TD-MPC and TD-MPC2 [13, 14] combine latent dynamics with value learning for continuous control. We study how a visual planner can accumulate predictive knowledge throughout deployment.

Predictive representations and JEPA. Prediction in learned representations is central to LeCun's proposed architecture for autonomous intelligence [17]. CPC [31] and SPR [29] learn representations through future prediction. DINO and DINOv2 [5, 25] provide transferable visual features, while R3M [24] learns manipulation features from human video. BYOL [7] predicts target-network embeddings; Barlow Twins [39] and VICReg [3] use redundancy reduction and statistical regularization to learn non-collapsed representations. I-JEPA [1] predicts target embeddings from image context, V-JEPA [4] extends feature prediction to video, and V-JEPA 2 [2] adds action-conditioned predictions for robotic planning. DINO-WM [40] predicts pretrained patch features and optimizes actions toward goal embeddings. Temporal straightening [35] improves planning geometry, DDP-WM [38] separates primary dynamics from background updates, and LeWorldModel [21] learns predictive embeddings end to end from pixels. SE-JEPA makes deployment experience persistent within this planning framework, consolidating unfamiliar conditions into

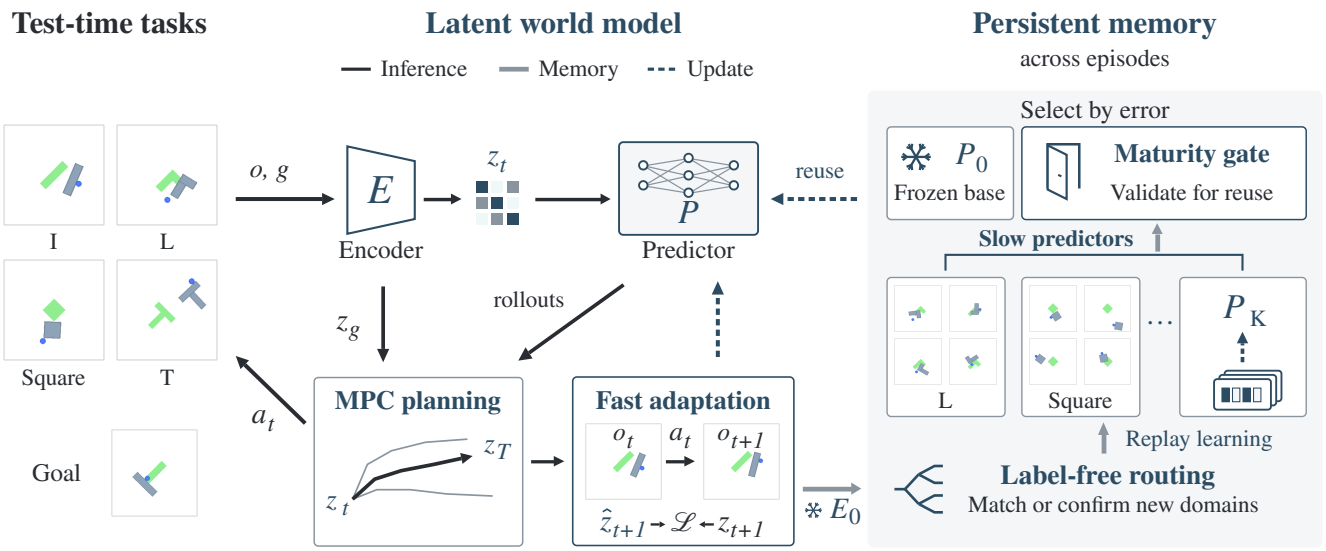


Figure 3. **SE-JEPA architecture.** The fast loop plans in latent space and adapts the predictor from observed transitions. At the end of an episode, the slow system re-encodes the trajectory with the frozen encoder, assigns unfamiliar experience to a self-discovered domain, and trains that domain's slow predictor by replay. Predictors that pass the maturity gate compete with the frozen base at the start of the next episode. Fast parameters reset at every episode; slow predictors and replay persist. Shape names illustrate memory contents and are not supervision.

predictors that can be retrieved on later encounters. We evaluate this mechanism with gradient-based and sampling-based action optimization.

Test-time adaptation. Test-time training [30] and entropy minimization [32] adapt deployed models using unlabeled target data; continual test-time adaptation follows a changing distribution [33]. Control supplies a natural self-supervised signal: every executed action yields an observation that checks the prediction. AdaJEPA [34] uses this signal inside MPC, with adaptation reset between episodes. SE-JEPA builds on this fast loop and retains its experience through slow consolidation, allowing subsequent episodes to start from a predictor informed by earlier interactions.

Continual learning and complementary memory. Complementary learning systems pair rapid hippocampal learning with slow neocortical consolidation [22]. Replay revisits earlier experience [26], GEM constrains updates using episodic memory [20], and EWC protects important parameters [16]. Learning without Forgetting preserves earlier outputs through distillation [18]; progressive networks retain separate task modules [27]. SE-JEPA brings complementary memory into predictive control, where retrieving a model changes both the actions taken and the data collected next. Its label-free routing organizes experience into domains, and its readiness gate evaluates consolidated predictors on episodes they have not yet trained on before using them for planning.

3. SE-JEPA

SE-JEPA couples episode-local adaptation with persistent predictive memory (Fig. 3). It keeps the pretrained encoder and predictor (E_0, P_0) as a fixed reference and learns additional predictors from deployment experience. Each discovered domain stores a slow predictor P_{ϕ_k}, a replay buffer $\mathcal{R}_k$, an update-direction prototype, and a validation history.

3.1. Deployment setting and motivation

An agent receives episodes $e = 1, 2, \ldots$, each with an initial observation o_0^e and a goal image g^e. It acts, observes the consequences, and may learn from them. Neither the type nor the onset of a distribution shift is announced; episode boundaries only mark when the scene is reset.

The world model consists of an encoder E_ψ and an action-conditioned predictor P_θ. With history length h, the encoder produces $z_t = E_\psi(o_t)$ and the predictor estimates $\hat{z}_{t+1} = P_\theta(x_t)$ from $x_t = (z_{t-h+1:t}, a_{t-h+1:t})$. The observation embedding includes visual features and proprioception. Rolling P_θ forward under a candidate action sequence, MPC solves

$$a^\star_{t:t+H-1} = \arg\min_{a_{t:t+H-1}} \mathcal{J}\big(\hat{z}_{t+1:t+H},\, E_\psi(g)\big), \qquad (1)$$

executes the first action block, observes the result, and replans. The optimizer is either GD or CEM. SE-JEPA leaves $\mathcal{J}$ and the optimizer unchanged; it changes only the predictor that MPC consults.

Table 1. **Keeping the adapted weights hurts.** Success on a matched 100-episode PushObj stream (I, square, T, I, square; 20 episodes each). Wins/losses count episodes solved only by the variant / only by reset.

Accumulation w	Success (%) ↑	Wins / losses
0 (episodic reset)	62.0	—
0.003	64.0	7 / 5
0.01	59.0	5 / 8
0.03	54.0	4 / 12
1 (keep all)	48.0	11 / 25

Why consolidate experience? Test-time adaptation produces a sequence of fast updates $\Delta_1, \Delta_2, \ldots$, one per episode. The direct inheritance rule accumulates a persistent offset, $s_{e+1} = s_e + w\Delta_e$, and starts the next episode from $\theta_0 + s_{e+1}$. Here $w = 0$ is episodic reset and $w = 1$ keeps every update in full.

Table 1 runs this rule on one fixed target order. Keeping every update lowers success by 14 points (exact McNemar test, $p = 0.029$), and small coefficients add at most two points (7 wins, 5 losses at $w = 0.003$). The update geometry offers a diagnostic clue. The mean cosine between consecutive updates is 0.061 on this stream and 0.018 over 2,449 consecutive pairs per planner in a longer seven-round stream (Appendix D). Each update solves a local problem, around one start state, one goal, and the trajectory the planner happened to take, and these local solutions point in nearly orthogonal directions. This motivates fitting the underlying transitions jointly rather than treating episode-specific parameter changes as reusable solutions.

We therefore consolidate the *transitions* behind fast adaptation into persistent predictors, grouping experience by predictive compatibility.

3.2. Fast adaptation and replay consolidation

Within an episode, the fast model follows AdaJEPA: after each replan, one Adam step updates the predictor's last block and encoder head using the recent observed transitions. These changes are reset at the next episode. A retrieved slow predictor can supply a better starting point for this adaptation.

To retain experience across episodes, we re-encode the executed trajectory with the frozen encoder E_0. This gives replay a stable coordinate system even as the fast encoder head changes. An accepted trajectory enters its domain's replay, and the slow predictor is trained on sampled transition windows:

$$\mathcal{L}_k(\phi_k) = \mathbb{E}_{(x,z^+)\sim\mathcal{R}_k} \left\| P_{\phi_k}(x) - z^+ \right\|_2^2. \tag{2}$$

Here x contains the latent history and executed actions, and z^+ is the next observation embedding. Sampling windows from earlier episodes fits their transitions jointly with the new experience. A new slow predictor starts from P_0; it learns from replay rather than inheriting the last fast update.

3.3. Discovering predictive domains

The reference model determines which episodes enter memory. We measure its newest-frame prediction error over the first three observed transitions and compare their geometric mean with a calibrated familiarity threshold. Episodes already explained by the reference use it as the base for fast adaptation and do not enter persistent replay. Slow learning is thus concentrated on experience that extends the reference model.

For other episodes, the fast parameter change Δ_e provides a direction fingerprint, $u_e = \Delta_e / \|\Delta_e\|_2$. An episode that retrieved a slow predictor first tries to write to that predictor's domain. Otherwise, cosine similarity to the stored direction prototypes proposes a domain, and a prediction-error check tests compatibility. Unmatched episodes wait in a pending pool. An incoming fingerprint aligned with two pending episodes establishes a new domain with those episodes. This discovers groups of compatible predictive experience without shape or appearance labels.

Domains can also share experience. We merge two domains when each predictor transfers to the other's recent episodes and both improve on the reference. Small immature fragments are absorbed by a mature domain that already predicts them well. Appendix A specifies the confirmation, merging, and absorption rules.

3.4. Validated retrieval for planning

Before training a slow predictor on an incoming episode, we evaluate its open-loop prediction of that episode's executed trajectory τ_e:

$$r_{e,k} = \frac{\mathcal{L}_{\text{roll}}(\phi_k; \tau_e)}{\mathcal{L}_{\text{roll}}(\theta_0; \tau_e)}. \tag{3}$$

This prospective measurement checks whether previous experience transfers to the new trajectory. A ratio above 1.2 rejects the proposed write. A predictor becomes eligible for retrieval when it has at least three validation measurements and their latest mean is at most 0.9.

During the first three replans, the reference and eligible slow predictors are compared on the transitions observed so far:

$$k^\star = \underset{k\in\{0\}\cup\mathcal{K}_{\text{eligible}}}{\arg\min} \mathcal{L}_{\text{pred}}(\phi_k; \mathcal{B}_{e,t}), \qquad \phi_0 = \theta_0. \tag{4}$$

The selected predictor supplies the base for fast adaptation, retaining the episode's current fast offset. Reference-familiar episodes restrict this choice to P_0. Prediction error therefore serves two distinct purposes: prospective validation determines which memories are eligible, and current observations determine which eligible memory to use.

Table 2. **Comparison with the state of the art (CEM).** Success in %; Steps is the mean number of MPC replans over all episodes, counting failures as 30. PushT Shift averages six visual shifts and Color the three recolorings. Rope and granular report Chamfer distance (CD). Rows above the rule use one test seed; their PushT targets are the established VIS50 set. *Italic*: published values from DINO-WM [40]. ^{T}: trained on T only. Rows below the rule use three test seeds on our paired held-out targets. Bold marks the largest success or smallest cost in each column; evaluation groups use different target sets.

	PushObj			PushT visual shifts			Rope	Granular
Method	All ↑	Unseen ↑	Steps ↓	Shift ↑	Color ↑	Steps ↓	CD ↓	CD ↓
R3M^{T} [24]	18.9	18.0	25.49	21.7	27.3	25.00	*1.13*	*0.95*
DINO CLS^{T}	14.9	10.0	26.77	34.7	35.3	22.38	*0.84*	*0.79*
DINO-WM [40]	62.0	46.0	16.15	—	—	—	0.491	0.301
LeWM [21]	64.0	50.0	16.28	40.3	16.0	21.45	0.786	0.322
DDP-WM [38]	57.7	43.3	17.59	52.7	57.3	18.39	0.677	0.259
Frozen	43.8	36.2	21.08	48.3	33.6	20.10	—	—
AdaJEPA [34]	70.4	54.4	16.02	60.2	42.2	18.46	0.629	0.304
SE-JEPA	**79.1**	**69.9**	**13.53**	**76.1**	**65.2**	**14.46**	**0.252**	**0.220**

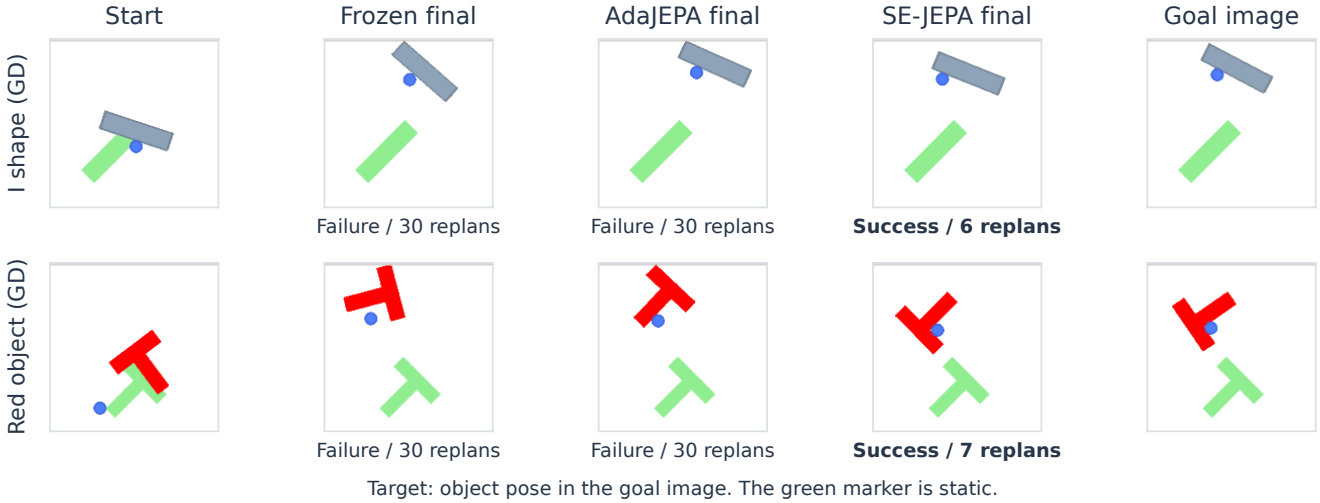


Figure 4. **Persistent memory resolves failures of episodic adaptation.** Selected GD examples on an unseen I shape (top) and red-object shift (bottom). Frozen and AdaJEPA exhaust 30 replans in both cases; SE-JEPA succeeds in six and seven, respectively. Each row shares the same start and goal. The target is the movable object's pose in the goal image; the green marker is static.

During deployment, replay, slow weights, and domain structure evolve after each episode. Held-out evaluation fixes this memory while retaining retrieval and episode-local adaptation, measuring how the accumulated experience transfers to new targets.

4. Experiments

4.1. Setup

Tasks. *PushObj* has seven object shapes. Our backbone is pretrained on T, L, Z, and plus; I, square, and small tee are unseen. *PushT* covers the original condition and six appearance shifts: blur, salt-and-pepper noise, darkness, and a red agent, object, or goal. *Rope* and *granular* manipulation use the deformable simulation of prior world-model work [19, 40].

Deployment and evaluation. For PushObj and PushT, SE-JEPA learns from streams of five rounds with 50 episodes per shape or condition, resampling targets every round, with three learning chains per planner. We then freeze each final memory and evaluate it on independent held-out targets: three test seeds with 50 episodes per shape or condition, whose source episodes never appear in the learning streams. For rope and granular, SE-JEPA consolidates a single slow predictor per task over four passes on 50 instances, planning with the slow predictor alone.

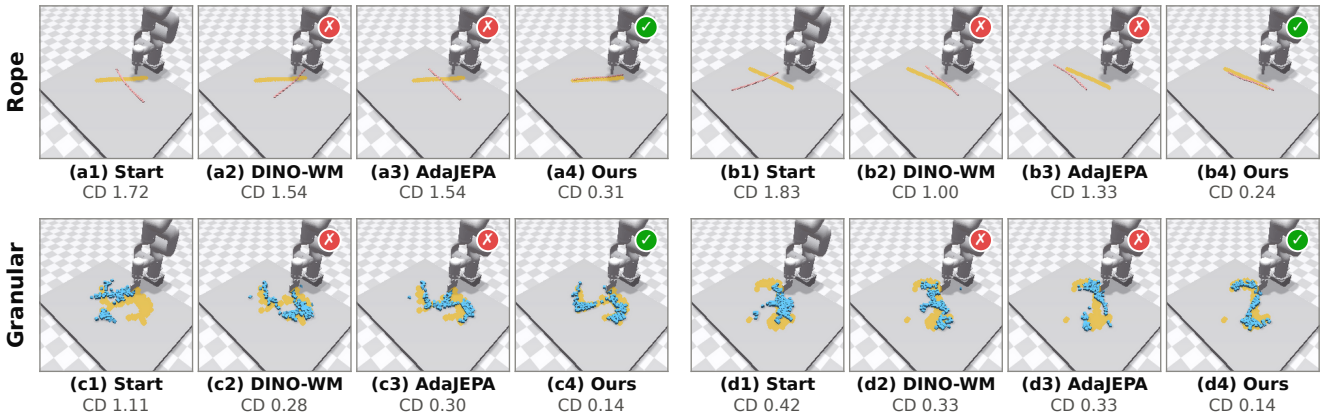


Figure 5. **Consolidated predictions support deformable manipulation.** Real FleX simulator trajectories for rope (top) and granular material (bottom), with two instances per task. Each group shows the start and the three methods' final states after 15 pushes; gold denotes the goal. These third-pass examples illustrate successful slow-only consolidation and were selected where both baselines fail. Frozen DINO-WM frames replay its executed actions. The aggregate fourth-pass results are reported in Table 2.

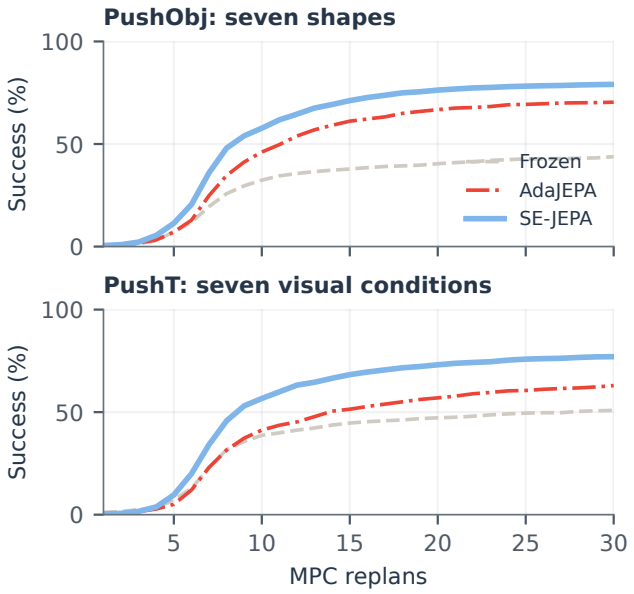


Figure 6. **Higher success within the same interaction budget.** Cumulative held-out success under CEM. Curves use all seven shapes (top) and all seven visual conditions (bottom).

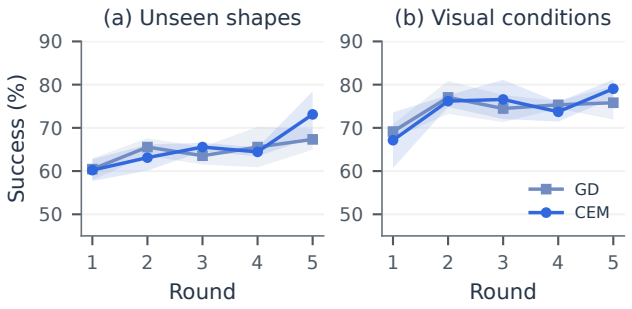


Figure 7. **Learning while deployed.** Online success of SE-JEPA over five rounds with targets resampled every round. Bands: variation across learning chains.

Baselines and planners. Frozen planning and AdaJEPA [34] isolate the effect of episodic adaptation; DINO-WM [40], LeWM [21], and DDP-WM [38] compare predictive world models, and DINOv2 CLS [25] and R3M [24] compare visual representations. Frozen and AdaJEPA use the released task-specific checkpoints. Appendix C records checkpoint and evaluation groups. MPC executes five actions per replan, up to 30 replans, with 100 GD steps or 10 CEM iterations of 200 samples and 30 elites.

4.2. Remembered experience improves held-out planning

Table 2 places the benefit of persistent memory in the context of recent world models. The paired comparison with AdaJEPA shows gains in both success and interaction efficiency. On PushObj it reaches 79.1% success, 8.7 points above AdaJEPA and 15 points above the best frozen world model, LeWM. The margin widens on unseen shapes, 69.9% against 54.4%, exactly where a fixed predictor is most wrong and remembered corrections help most. SE-JEPA also finishes faster: 13.53 replans on average against 16.02 for AdaJEPA, a 16% reduction in interaction.

The visual shifts show the clearest gap. Averaged over six shifts, SE-JEPA reaches 76.1% against 60.2% for AdaJEPA. The largest gains over episodic adaptation occur under object and goal recoloring; SE-JEPA raises red-object success from 14.0% to 47.1% and red-goal success from 41.3% to 68.9%. It improves on all seven conditions, in-

cluding the original one (Appendix D).

The gains hold for gradient-based planning. With GD, SE-JEPA reaches 81.6% on PushObj (AdaJEPA: 72.2%) and 72.4% on unseen shapes (55.3%), and 75.7% across the seven visual conditions (58.7%), while cutting mean replans from 14.52 to 11.30 and from 17.96 to 13.21.

On deformable objects, SE-JEPA reduces Chamfer distance to 0.252 on rope and 0.220 on granular material. This is 49% and 27% below the DINO-WM model it builds on, with the lowest reported CD among the methods in Table 2. Fast adaptation is switched off in this setting, so the improvement comes entirely from consolidated knowledge.

Remembered experience also makes each interaction more productive (Fig. 6). The cumulative success curves show how the benefit develops across the planning budget, complementing the mean replan counts in Table 2.

4.3. Qualitative results

Figure 4 contrasts selected episodes in which persistent memory changes the outcome. On the unseen I shape, SE-JEPA reaches the goal in six replans; under object recoloring, it succeeds in seven. Frozen and AdaJEPA fail within the 30-replan budget in both cases. The red-object example makes the difference visible: the baselines leave the T-shaped object at the wrong orientation, whereas SE-JEPA brings it into the goal pose. Additional complete trajectories appear in Appendix E.

Figure 5 extends the behavioral evidence to deformable objects. The consolidated predictor brings the rope and particles toward the target configuration using slow memory without within-episode adaptation. This complements the aggregate CD reductions: deployment experience improves the physical outcomes that the planner reaches.

4.4. Stronger with deployment

Figure 7 tracks knowledge accumulation during deployment, with new targets in every round. With CEM, unseen-shape success rises from 60.2% in the first round to 73.1% in the fifth, and success over the seven visual conditions from 67.1% to 79.0%. GD follows the same trend, from 60.4% to 67.3% and from 69.1% to 75.8%.

Deformable manipulation shows the same behavior (Fig. 9). Rope CD falls from 0.469 after the first pass to 0.252 after the fourth, and granular CD from 0.257 to 0.220, with both curves well below the pretrained DINO-WM. The knowledge also transfers to new instances: a rope predictor saved after the first pass, frozen and tested on 91 unseen instances, lowers CD from 0.664 to 0.349.

5. Conclusion

SE-JEPA makes deployment experience a lasting resource for latent planning. Fast adaptation handles the current episode; replay consolidates its transitions into predictors for self-discovered domains; prospective validation determines when those predictors enter the planning loop. This combination improves held-out success and reduces interactions on shape and visual shifts with both GD and CEM. Slow consolidation also improves rope and granular manipulation, demonstrating the value of persistent predictive memory for deformable dynamics. By retaining a fixed pretrained reference alongside learned experience, SE-JEPA provides a practical foundation for world models that continue learning through use.

Future work. Our label-free gate groups experience by predictive compatibility, which need not fully distinguish tasks with similar short-term dynamics. Improving task discrimination with longer interaction histories is a promising direction for more reliable routing and memory reuse across diverse tasks.

LLM Usage Statement

Large language models (OpenAI Codex and Anthropic Claude) were used to draft and revise substantial portions of the manuscript text, and to assist with literature and source checks, LaTeX organization, figure generation, and result aggregation. They also assisted with experimental code, including offline world-model adapters and experiment orchestration. All quantitative results are taken from saved experimental outputs or explicitly cited published sources; language-model output is not used as experimental evidence. The authors take full responsibility for the claims, citations, code, and final verification of the manuscript.

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Supplementary Material

A. Implementation and Memory Management

Compute environment. Model training and evaluation were conducted on systems equipped with 8× NVIDIA GeForce RTX 5090 GPUs and 8× NVIDIA A100 GPUs.

A.1. Fast and slow learning

The label-free rigid-object configuration builds on the episodic adaptation loop of AdaJEPA [34]. At every replan, the most recent observed transitions provide a self-supervised next-latent loss. The predictor's last block and the encoder's designated projection/head parameters are adapted with one Adam step. The episode-local predictor and encoder changes are reset at the next episode. A retrieved slow predictor supplies the predictor initialization; it does not replace the frozen reference encoder.

Persistent replay is re-encoded by the reference encoder. Each slow learner is a full predictor copy, initialized from the pretrained predictor rather than from the final fast parameters of its founding episode. For each slow optimization step, up to eight stored episodes are sampled with replacement; each contributes a random window of up to six latent frames. The prediction objective averages one-step observation-feature errors over the available history windows. Action features condition prediction but are not reconstruction targets. Standard writes use 15 Adam steps. Founding a learner after n stored episodes uses $15n$ steps to process the accumulated evidence. Merging likewise adds replay updates for absorbed experience.

Replay stores compact latent trajectories rather than images. The pending pool is bounded, and merging and absorption remove redundant predictor copies as compatible domains are recognized.

A.2. Order of decisions

Algorithm 1 summarizes a learning episode. Familiarity uses the geometric mean of newest-frame prediction errors on up to the first three observed transitions. The threshold is calibrated once on the shape backbone and transfers unchanged to the visual experiments.

The selected predictor's error is evaluated before it learns from the current episode, so maturity reflects transfer to data the predictor has not fitted. In read-only evaluation, the episode-end write, validation-history update, slow training, merging, and absorption steps are all disabled.

A.3. Fingerprints and delayed domain founding

A fingerprint concatenates the adapted predictor and encoder-head changes and is normalized to unit length. Domain prototypes accumulate normalized fingerprints from episodes planned from the reference; episodes starting from a slow predictor do not redefine that reference-based prototype. If an episode has already read a learner, it preferentially writes to that learner's domain unless the prospective error ratio rejects the write. Otherwise, it tries compatible domain directions in descending cosine order.

An unassigned episode enters a pending pool. It confirms a new domain when at least two pending fingerprints align with its own above the new-domain threshold. These members need not be pairwise aligned with one another; the incoming fingerprint is the common reference for this confirmation. The members and the new episode are transferred together to the new replay. No ground-truth shape or appearance category participates in this procedure.

Table 3. Label-free configuration used for PushObj and PushT. The seven-round stream in Appendix D uses a slow learning rate of 2×10^{-4}.

Setting	Value
Fast predictor learning rate	5×10^{-4}
Fast encoder-head learning rate	10^{-5}
Fast steps per replan	1
Recent adaptation buffer	5 transitions
Slow learning rate	10^{-4}
Slow steps per standard write	15
Slow batch size (episodes)	8
Slow sampled window (frames)	up to 6
Familiarity threshold τ_f	0.0146
Existing-domain cosine threshold	0.10
New-domain cosine threshold	0.05
Episodes to confirm a domain	3
Pending-pool capacity	40
Minimum episodes to train a learner	3
Retrieval window	first 3 replans
Maturity history length m	3
Maturity error-ratio threshold τ_m	0.9
Write rejection ratio	1.2
Merge evidence per direction	2 episodes
Merge relative tolerance	0.5
Fragment absorption interval	25 episodes
Absorption error-ratio threshold	0.8
Fragment-size limit	< 6 episodes

A.4. Merging and fragment absorption

For a new unfamiliar episode, every existing trained learner is evaluated before training. The resulting ratios provide cross-domain evidence. Let $G_{A\to B}$ be the geometric mean error ratio of learner B on recent episodes assigned to A, and $G_{A\to A}$ the corresponding ratio for learner A. Domains merge only if both directions satisfy

$$G_{A\to B} \le \min\{1, (1+\epsilon_m) G_{A\to A}\}, \tag{5}$$

Algorithm 1 One deployment-learning episode of SE-JEPA

Require: Frozen (E_0, P_0), memory $\mathcal{M}$, initial observation o, goal g
Reset fast parameters to (E_0, P_0)
Initialize recent buffer $\mathcal{B}$ and episode trajectory τ
while budget remains and task is not complete **do**
    Optimize an action sequence with the current fast model
    Execute its first block; observe the next state
    Append real transitions to $\mathcal{B}$ and τ
    if within the early retrieval window **then**
        Form candidates: P_0 and mature slow predictors
        Restrict to P_0 if the observed episode is familiar
        Select the smallest observed prediction error
        Load the selected predictor, retaining the fast offset
    end if
    Adapt the fast model on recent real transitions
end while
Encode τ with frozen E_0
Measure pre-update rollout errors of existing predictors
if episode is unfamiliar **then**
    Propose a domain using retrieval or an update fingerprint
    Reject an incompatible write, or retain pending evidence
    if a domain is assigned or newly confirmed **then**
        Record prospective validation when a learner exists
        Append latent experience; update its slow predictor
        Check bidirectional merges and periodic absorption
    end if
end if
Discard episode-local fast parameters

with the symmetric condition for $B \to A$. Each direction requires two jointly evaluated episodes in the primary configuration. The larger domain retains its learner and incorporates the union of stored experience. Merged identifiers are not reused.

Every 25 episodes, small immature domains can be absorbed into a mature predictor that explains their experience with an error ratio at most 0.8. The primary configuration limits absorption to fragments with fewer than six episodes and fewer than three maturity measurements. Pending-pool absorption and the additional absolute-error gate are disabled in the shape and visual experiments.

B. Evaluation Protocols

B.1. Shape shifts

The shape learning stream has five deployment rounds over seven shapes, with 50 episodes per shape in every round. The held-out benchmark uses target seeds 2000, 2100, and 2200, each with 350 episodes. I, square, and small tee are unseen in pretraining; T, L, Z, and plus are seen.

Each planner has three learned memory states. Every state is evaluated on all three held-out seeds; for each seed we average over episodes and states, and standard deviations are computed over the three seed-level means. The CEM states are S7t3, S7t3b, and S7t3s700, where S7t3b is a rerun of S7t3 that replaces a run in which no learner matured. Frozen and AdaJEPA use the same three held-out seeds; the external baselines use seed 2000.

B.2. Visual shifts

The paired visual evaluation contains the original T condition, blur, salt-and-pepper noise, darkness, red agent, red object, and red goal. Three five-round learning chains resample targets every round, with 350 episodes per round. Held-out testing uses three target seeds and 50 episodes per condition. Direct checks of source-episode identifiers confirm that held-out sources never occur in the learning streams and that the three test seeds use disjoint sources.

The external visual baselines use the established VIS50 collection with planning seed 100. The external comparisons use its six shifted conditions, with 50 paired initial/goal states per condition.

B.3. Success and interaction counts

Rigid-object success follows the environment's goal test: object-position error below 20 simulator units and angular error below $\pi/9$. Each executed model action contains five physical actions, and an episode has at most 30 MPC replans. For episode i with success indicator S_i and first successful replan T_i,

$$\overline{T} = \frac{1}{N}\sum_{i=1}^{N}\left[\, S_i T_i + (1 - S_i)\, T_{\max}\right], \qquad T_{\max} = 30. \quad (6)$$

Equivalently, $\overline{T} = T_{\max} - \sum_{t=1}^{T_{\max}-1} F(t)$, where $F(t)$ is the fraction of episodes solved by replan t; the cumulative curves in Fig. 6 visualize F.

B.4. Deformable manipulation

Rope and granular each use one slow predictor, a DINO-WM backbone, and 50 instances revisited over four passes, with the same targets, order, and planning seeds in every pass. The slow-only variant plans from the slow predictor without fast within-episode updates. CEM uses 300

samples, 30 elites, and 10 iterations, with three MPC iterations of five pushes each. The unseen-rope evaluation freezes the predictor saved after the first pass and tests it on 91 instances absent from the learning stream, against the frozen reference on the same instances. Two frozen granular episodes with unstable batched particle simulation were rerun individually with identical settings and seeds.

C. Baseline Checkpoints and Comparisons

Predictive world models. The comparisons focus on latent world models used for visual planning: DINO-WM, LeWM, and DDP-WM, with DINO CLS and R3M as representation baselines. Frozen and AdaJEPA use released task-specific checkpoints. The shape backbone of SE-JEPA is trained for three epochs on 16,000 trajectories covering T, L, Z, and plus. The additional DINO-WM shape model uses the same training trajectory count and epoch budget, with 4,000 validation trajectories and batch size 32. DINO CLS and R3M use existing T-only checkpoints; LeWM and DDP-WM use their corresponding checkpoint adapters.

Evaluation groups. The paired Frozen/AdaJEPA/SE-JEPA comparison uses the independent held-out target sets in Appendix B. External visual baselines use VIS50, with 50 initial/goal pairs in each of six shifted conditions and planning seed 100. The table separates these evaluation groups. Within each group, methods receive the same targets and planning budget. PushT Shift averages blur, noise, dark, red agent, red object, and red goal; Color averages the last three. The original condition is included in the seven-condition tables and curves, but excluded from Shift.

Deformable results. DINO-WM is the frozen backbone for the rope and granular experiments. AdaJEPA adapts within each episode, whereas SE-JEPA uses a consolidated slow predictor. Italic entries for R3M and DINO CLS are published values from DINO-WM [40]; the other entries are measured results. The repeated-instance and unseen-rope protocols are specified in Appendix B.

D. Additional Results

D.1. All object shapes

Table 5 reports every held-out shape. SE-JEPA gives the largest gains on the unseen shapes: with GD, I rises from 44.7% (AdaJEPA) to 77.6% and square from 46.7% to 61.1%. It also leads on the seen shapes in aggregate.

D.2. All visual conditions

Table 6 and Fig. 8 break the paired visual results down by condition. SE-JEPA improves on all seven conditions with both planners. The largest gains, more than 25 points, appear under red-object and red-goal recoloring, where episodic adaptation has to rediscover the new appearance in every episode.

D.3. Planning efficiency

Figure 6 plots cumulative held-out success against the number of MPC replans. SE-JEPA pulls ahead within the first four replans and keeps the lead for the rest of the budget, showing how the advantage develops before the final 30-replan evaluation.

D.4. Deformable manipulation

Table 7 adds standard errors to the deformable results of Table 2. Table 8 follows consolidation pass by pass at two slow learning rates. With 10^{-4}, CD decreases at every pass on both tasks. With 2×10^{-4}, rope improves faster in the early passes and both rates reach a similar level after four passes, so consolidation is robust to the learning rate.

D.5. A seven-round shape stream

A second PushObj stream runs seven rounds with a slow learning rate of 2×10^{-4}, one chain per planner. Each round has 50 episodes per shape, drawn without replacement and excluding the held-out sources, for 2,450 episodes per planner. Online success rises from 70.6% in the first round to 78.6% in the seventh with GD, and from 72.3% to 78.3% with CEM.

The episode logs record the fast-update fingerprint of every episode. Figure 10 shows the cosine between consecutive fingerprints: the mean is 0.018 for both planners, and most of the mass lies near zero. This is the geometry behind the failure of weight accumulation in Sec. 3.1. Over the stream, SE-JEPA writes 622 (GD) and 663 (CEM) episodes to memory, 25.4% and 27.1% of all episodes, and finishes 630 and 666 episodes with a slow predictor.

D.6. What SE-JEPA chooses to learn

SE-JEPA learns selectively. In a seven-round PushObj stream of 2,450 episodes per planner, it writes only 25% (GD) to 27% (CEM) of episodes to memory; the rest are familiar to the pretrained model, still pending, or rejected by validation. Retrieval is just as targeted: slow predictors plan about one episode in four, and the reference handles the rest.

Selective writing leaves familiar dynamics to the pretrained model, which SE-JEPA never overwrites. On the four pretraining shapes, SE-JEPA reaches 86.0% held-out success with CEM and 88.4% with GD, against 82.3% and 84.8% for AdaJEPA. This preserves a useful pretrained reference while allowing specialized predictors to add capacity for recurring shifts.

Table 4. Both planners on the paired held-out sets. Shape All and Unseen use held-out PushObj targets. Visual All averages seven conditions and Shift six; visual Steps averages all seven conditions.

		PushObj			PushT		
Planner	Method	All ↑	Unseen ↑	Steps ↓	All ↑	Shift ↑	Steps ↓
GD	Frozen	42.48	32.44	20.02	44.10	41.56	19.94
GD	AdaJEPA	72.19	55.33	14.52	58.67	54.89	17.96
GD	SE-JEPA	81.59	72.44	11.30	75.68	73.63	13.21
CEM	Frozen	43.81	36.22	21.08	50.86	48.33	19.56
CEM	AdaJEPA	70.38	54.44	16.02	62.95	60.22	17.87
CEM	SE-JEPA	79.08	69.85	13.53	77.05	76.11	14.15

Table 5. Held-out success (%) per PushObj shape. I, square, and small tee are unseen in pretraining. All averages the seven shapes.

Planner	Method	I	L	Square	Z	Small tee	Plus	T	All
CEM	Frozen	32.7	52.7	23.3	41.3	52.7	43.3	60.7	43.8
CEM	AdaJEPA	49.3	82.0	44.7	82.0	69.3	79.3	86.0	70.4
CEM	SE-JEPA	72.7	84.2	58.4	86.2	78.4	85.6	88.0	79.1
GD	Frozen	24.0	52.7	28.0	40.0	45.3	48.7	58.7	42.5
GD	AdaJEPA	44.7	90.0	46.7	82.0	74.7	82.0	85.3	72.2
GD	SE-JEPA	77.6	88.4	61.1	88.0	78.7	87.6	89.8	81.6

E. Qualitative Examples and Reproducibility

E.1. Full trajectories

Figure 4 selects an unseen-I and a red-object episode where both baselines fail and SE-JEPA succeeds under GD. These illustrate changes in planning behavior; aggregate success is evaluated on the complete held-out sets. The source package supplies the first held-out target of test seed 2000 for each condition, and the displayed conditions are chosen by outcome. All methods share identical initial and goal observations. The panels use a consistent crop size to enlarge the movable objects, with every colored foreground element preserved.

Figures 11 and 12 provide additional CEM examples on the first held-out target of the same test seed: an unseen square and a red-agent shift. They show the complete trajectories and each method's prediction loss. These are separate from the GD examples in the main paper.

Each loss curve is measured in the latent space of its own model, so it describes how prediction error evolves within a method over the episode. Task success, judged in the shared simulator state, is the common yardstick across methods.

E.2. Reproducibility

Tables and plots use the bundled reporting snapshot and the archived Drive experiment packages. The revised trajectory panels are rendered directly from the packages' simulator frames; the asset manifest records their source files and hashes. Success and Steps are computed from the final episode files and their run configurations. Offline world-model checkpoints record the upstream code revision, the epoch budget, the model configuration, and hashes of the adapter sources. Code, configurations, and learned memory states will be released.

Table 6. Held-out visual success (%) per condition on the paired targets. All averages seven conditions; Shift excludes Original.

Planner	Method	Original	Blur	Noise	Dark	Red agent	Red object	Red goal	All	Shift
CEM	Frozen	66.0	56.0	62.7	70.7	50.0	14.0	36.7	50.9	48.3
CEM	AdaJEPA	79.3	78.0	76.7	80.0	71.3	14.0	41.3	63.0	60.2
CEM	SE-JEPA	82.7	86.4	86.7	88.0	79.6	47.1	68.9	77.0	76.1
GD	Frozen	59.3	48.7	55.3	58.7	50.7	6.7	29.3	44.1	41.6
GD	AdaJEPA	81.3	77.3	73.3	73.3	58.0	10.7	36.7	58.7	54.9
GD	SE-JEPA	88.0	88.7	84.9	87.3	74.7	44.4	61.8	75.7	73.6

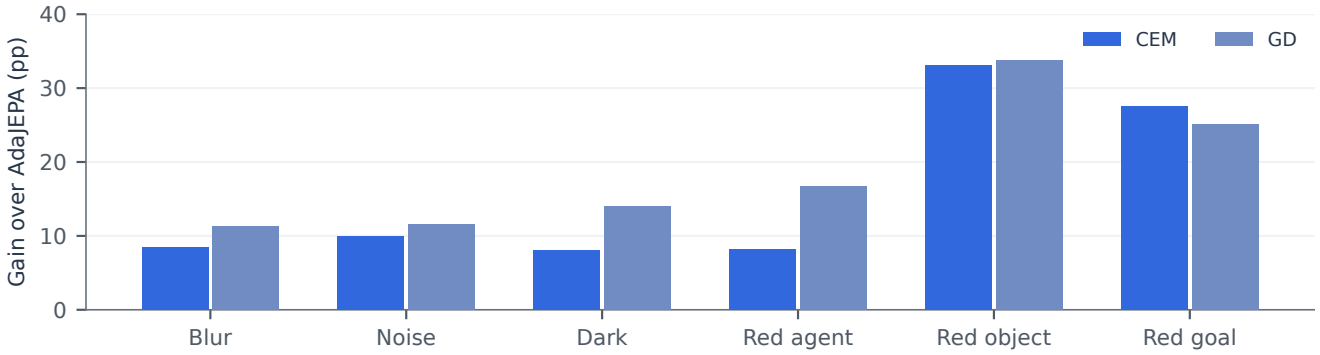


Figure 8. Held-out gain of SE-JEPA over AdaJEPA for each visual shift, in percentage points.

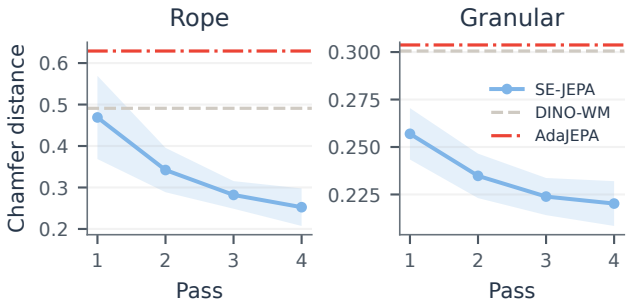


Figure 9. **Consolidation on deformable objects.** Chamfer distance of SE-JEPA (slow predictor only) over four passes on 50 instances; bands show standard error. Lines mark DINO-WM and AdaJEPA.

Table 7. Deformable manipulation: mean CD $\pm$ standard error over 50 instances. SE-JEPA uses its slow predictor after four passes.

Method	Rope CD ↓	Granular CD ↓
DINO-WM	0.491 ± 0.059	0.301 ± 0.020
LeWM	0.786 ± 0.112	0.322 ± 0.018
DDP-WM	0.677 ± 0.086	0.259 ± 0.012
AdaJEPA	0.629 ± 0.096	0.304 ± 0.019
SE-JEPA	$\mathbf{0.252 \pm 0.045}$	$\mathbf{0.220 \pm 0.012}$

Table 8. CD of SE-JEPA after each consolidation pass at two slow learning rates (mean $\pm$ standard error, 50 instances).

	lr 10^{-4}		lr 2×10^{-4}	
Pass	Rope	Granular	Rope	Granular
1	0.469 ± 0.100	0.257 ± 0.014	0.416 ± 0.070	0.260 ± 0.012
2	0.342 ± 0.054	0.235 ± 0.012	0.286 ± 0.050	0.241 ± 0.012
3	0.282 ± 0.033	0.224 ± 0.010	0.283 ± 0.037	0.247 ± 0.012
4	0.252 ± 0.045	0.220 ± 0.012	0.243 ± 0.033	0.243 ± 0.013

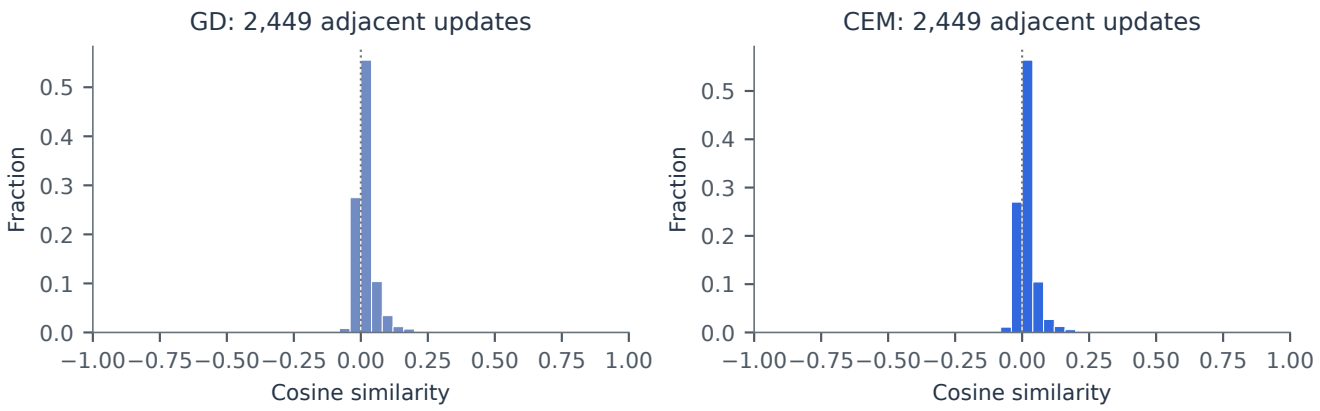


Figure 10. Cosine similarity between consecutive fast-update fingerprints in the seven-round shape stream (2,449 pairs per planner).

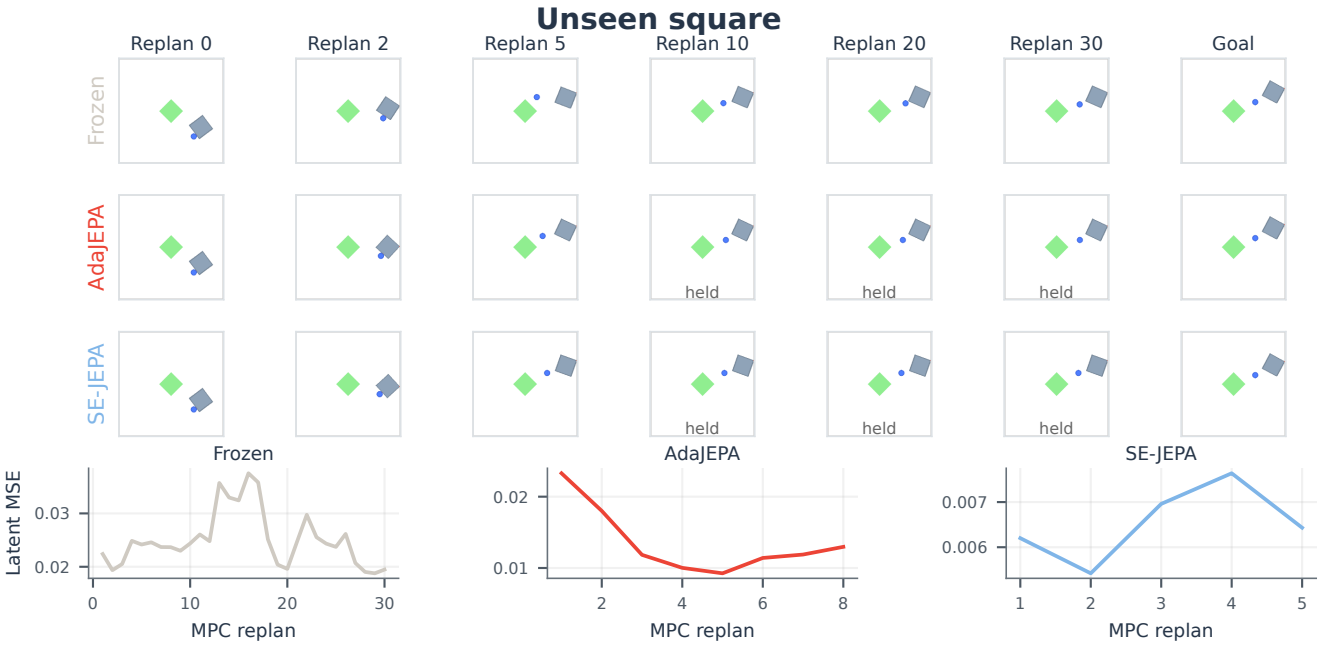


Figure 11. Full unseen-square example with intermediate frames and prediction-loss traces. Frames marked held repeat the terminal state after success. SE-JEPA reaches the goal at replan five, AdaJEPA at replan eight, and the frozen model does not succeed within the budget.

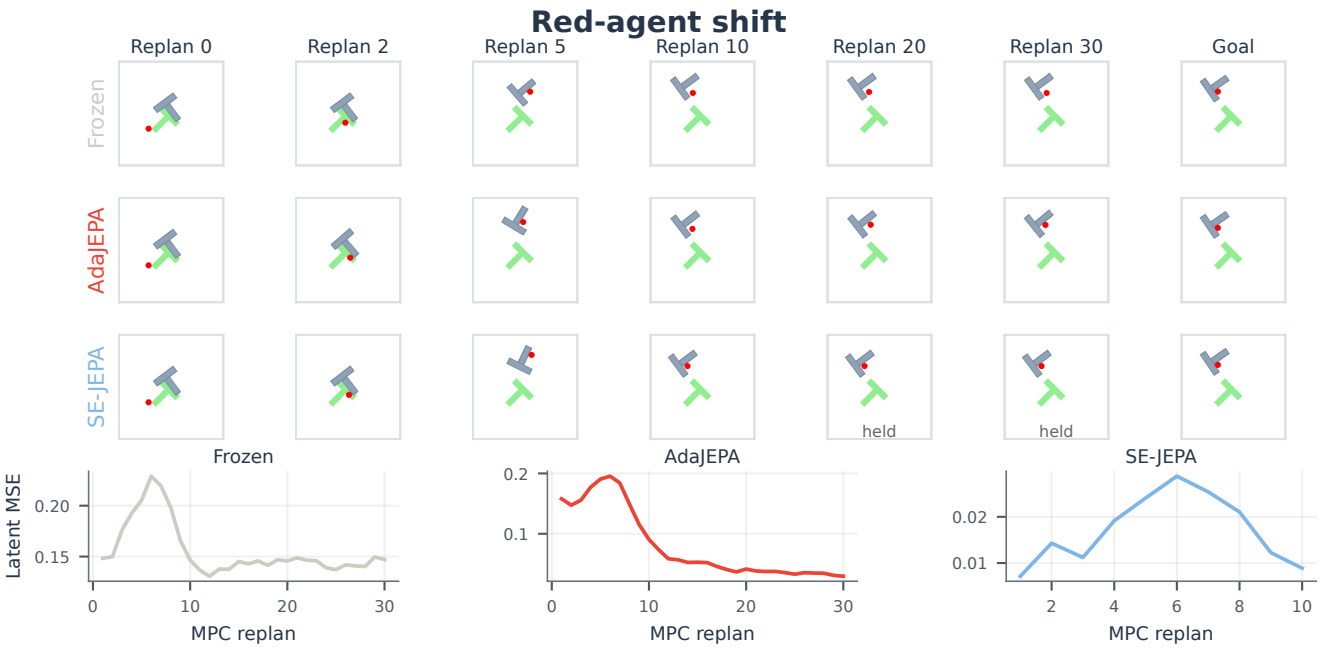


Figure 12. Full red-agent example, replayed in the original simulator from the executed actions. SE-JEPA succeeds at replan ten; Frozen and AdaJEPA reach the 30-replan limit.